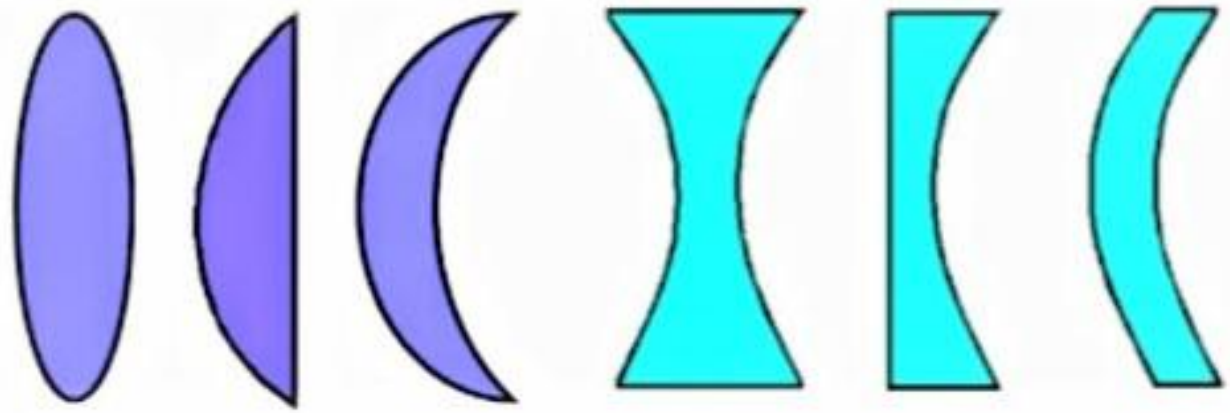
Fizika

9-sinf

Mavzu: **Linzalar.**

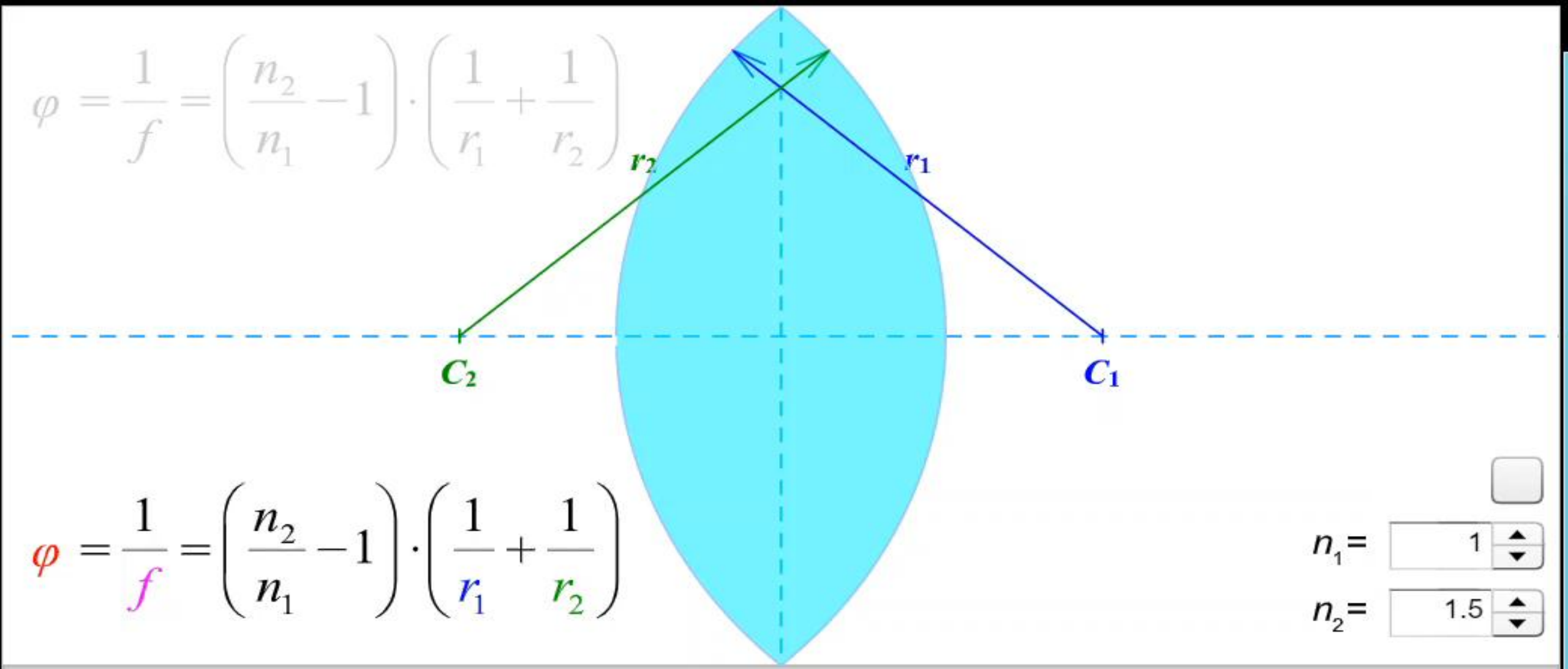


Qvariq va botiq linzalar



Bir yoki ikki tomoni sferik sirt bilan chegaralangan shaffof jism linza deb ataladi. Linzalar qavariq va botiq bo‘ladi.

$$\varphi = \frac{1}{f} = \left(\frac{n_2}{n_1} - 1\right) \cdot \left(\frac{1}{r_1} + \frac{1}{r_2}\right)$$



$$\varphi = \frac{1}{f} = \left(\frac{n_2}{n_1} - 1\right) \cdot \left(\frac{1}{r_1} + \frac{1}{r_2}\right)$$

n_1= 1

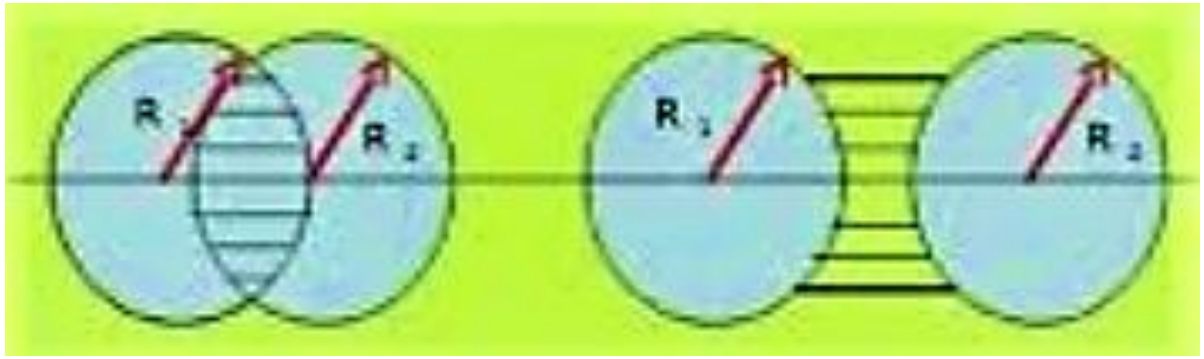
n_2= 1.5

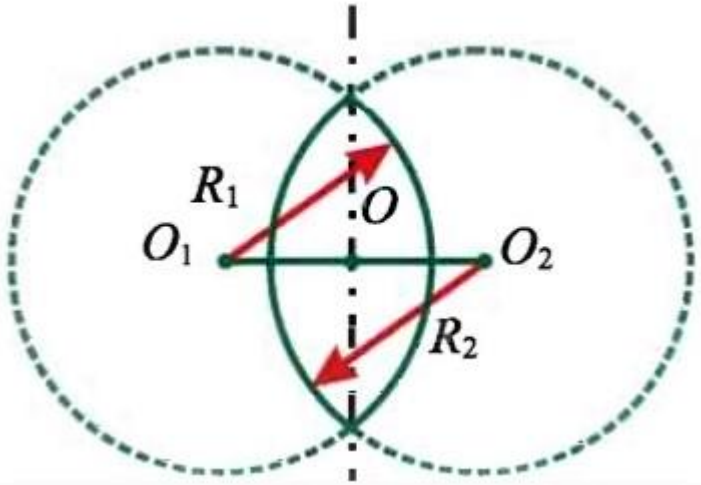
r_1 = +20 cm

r_2 = +20 cm

f = 0.2 m

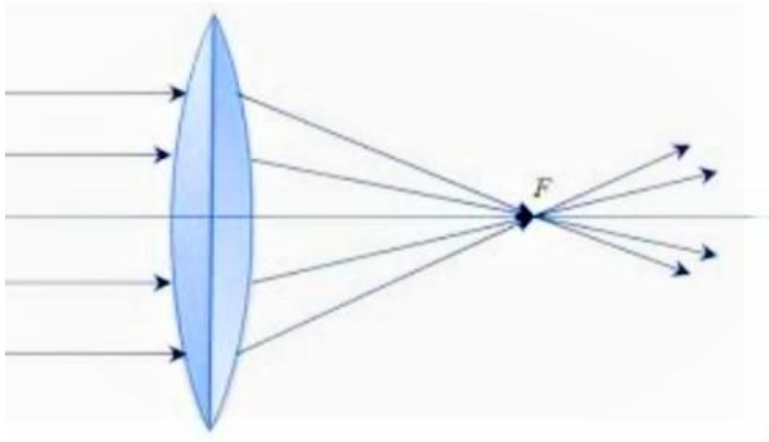
φ = 5 D

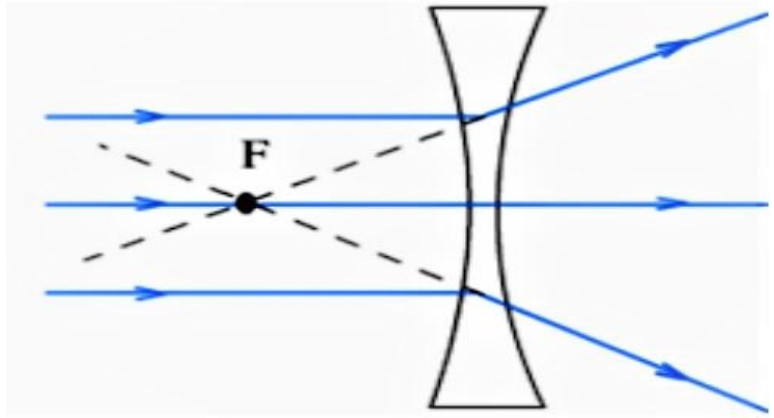




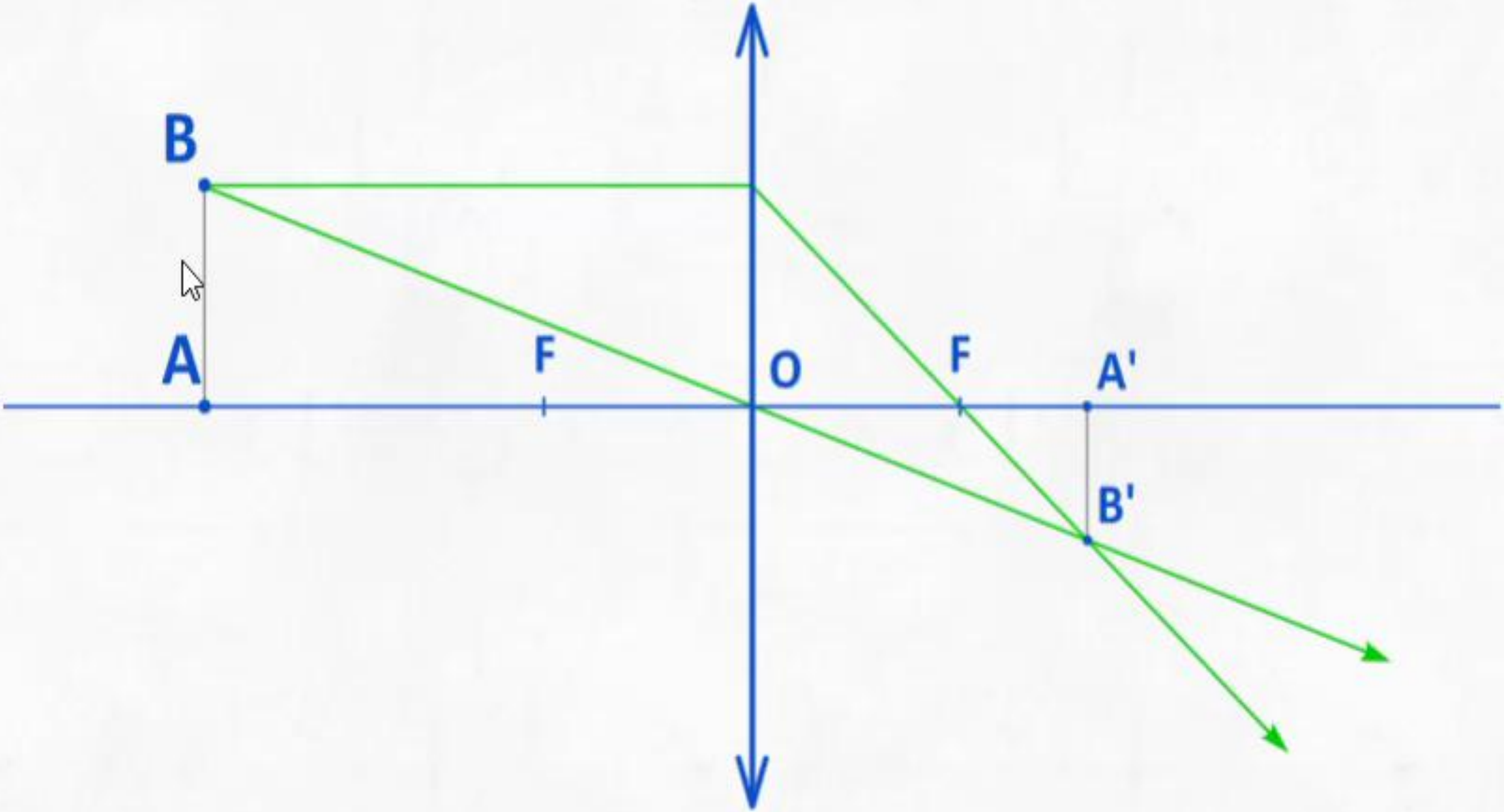
Linzalar $\boldsymbol{R_1}$ va $\boldsymbol{R_2}$ radiusli sferalarning o‘zaro kesishishidan hosil bo‘ladi. O nuqta linzaning ***markazi***, O_1O_2 to‘g‘ri chiziq linzaning ***bosh optik o‘qi*** deyiladi.

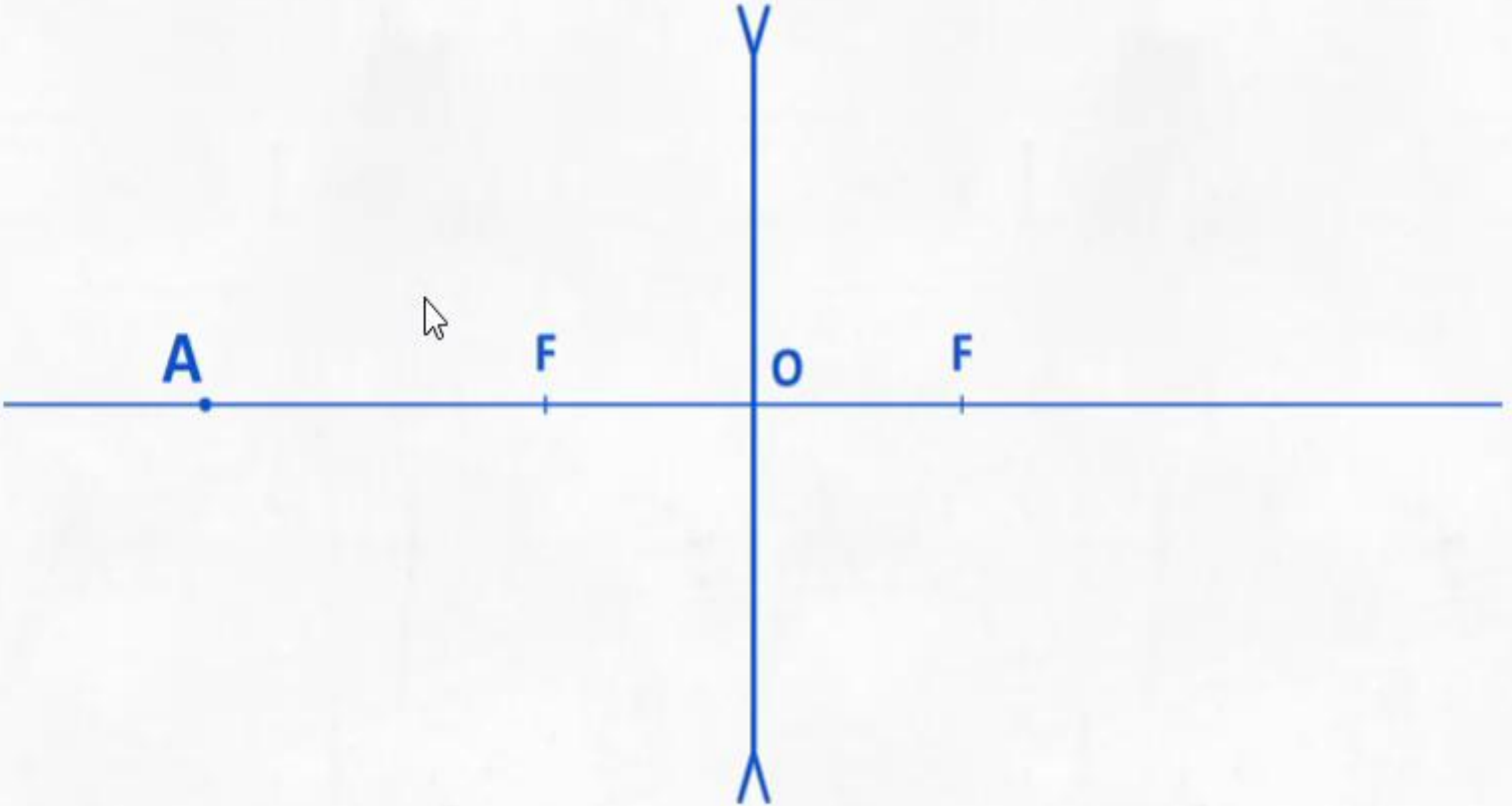
Yig‘uvchi va sochuvchi linza





Agar qavariq linzaga bosh optik o‘qiga parallel bo‘lgan nurlar yo‘naltirsak, linzadan o‘tgach bir nuqtada yig‘iladi. Shuning uchun qavariq linza ***yig‘uvchi linza*** deb ataladi. Botiq linzaga parallel nurlar yo‘naltirsak, nurlar linzadan o‘tgach bir tekis sochiladi. Shuning uchun botiq linza ***sochuvchi linza*** deb ataladi.


B
A
F
O
F
A'
B'


A
F
O
F

Linzaning fokusi va optik kuchi

Fokus masofasiga teskari bo‘lgan kattalik linzaning optik kuchi deyiladi va $\boldsymbol{D}$ harfi bilan belgilanadi.

$$\boldsymbol{D = \frac{1}{F}} \qquad \boldsymbol{F = \frac{1}{D}}$$

$$\boldsymbol{[D] = \frac{1}{m} = 1\ \mathrm{dptr}(\mathrm{diopteriya})}$$

$$\boldsymbol{F = \frac{1}{(n-1)\cdot\left(\frac{1}{R_1}+\frac{1}{R_2}\right)}}$$

Masala

Fokus masofasi $20\ cm$ bo‘lgan linzaning optik kuchi qancha?

Fokus masofasi $-10\ cm$ bo‘lgan linzanikichi?

Berilgan:

$1)\ F = 20\ cm = 0{,}2\ m$

$2)\ F = -10\ cm = -0{,}1\ \text{m}$

Topish kerak: $D - ?$

Formula va yechish:

$$D = \frac{1}{F}$$

$$1)\ D = \frac{1}{0{,}2\ m} = 5\ \text{dptr}$$

$$2)\ D = \frac{1}{(-0{,}1\ m)} = -10\ \text{dptr}$$

Javob: **1)** $\boldsymbol{D = 5\ dptr}$;

2) $\boldsymbol{D = -10\ dptr}$.

Mustaqil bajarish uchun topshiriqlar

1. Mavzuni o‘qish va savollarga javob yozish.(145-bet)

2. 27-mashqning 1-masalasini yechish.(149-bet)